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Reading the Reader — User Guide

A handbook for researchers operating the Reading the Reader adaptive reading platform.

1. Overview

Reading the Reader is a **researcher-operated** adaptive reading system. It connects to a Tobii eye tracker, runs a controlled reading session for an invited participant, mirrors the participant's screen for the researcher in real time, and can apply context-aware micro-interventions while the participant reads. After a session, all data can be exported or replayed for analysis.

A typical experiment moves through these stages:

1. **Prepare content** — save one or more reading texts as reusable *reading materials*.
2. **Compose a template** — combine materials into a reusable *experiment template* with presentation defaults and a runtime strategy.
3. **Run a session** — follow the guided stepper to set up the tracker, baseline, participant details, and calibration, then start reading.
4. **Monitor live** — watch the participant's gaze, attention, and struggle signals; trigger or approve interventions.
5. **Export / replay** — save the recorded session and review it later.

2. Before You Start

Requirement	Detail
Browser	A Chromium-based browser is recommended. Fullscreen permission is required for calibration and the live view.

Requirement	Detail
Eye tracker	A Tobii device connected to the Windows machine (for real eye-tracking sessions).
Backend running	The backend service must be running and reachable by the frontend.
Two screens (recommended)	One screen for the Participant View , one for the Researcher Live View .
Content format	Reading materials are Markdown only . PDF is not supported.

Tip: If you do not have hardware available, switch to **Mouse mode** in Settings to run a demo where the participant's mouse position acts as synthetic gaze. See [Sensing Modes](#).

3. Tour of the Workspace

The workspace is where the researcher prepares, runs, and reviews experiments. The left **sidebar** is the primary navigation the researcher uses to move between the reading-material library, experiment templates, the live session view, replay, and settings. Each sidebar item is described below.

Sidebar item	What it's for
Home	The researcher dashboard. Lists "Ready to run" templates and drafts; start or edit a template from here.
Material Library	Create, edit, export, and delete reusable reading materials.
Experiment Templates	Manage reusable experiment setups (ready / drafts / archived). Duplicate, export, edit, delete.
Researcher Live	The live monitoring view of the currently running session.
Participant View	Opens the participant-facing flow (opens in a new tab — put this on the participant's screen).
Replay	Load a saved or exported session and play it back frame by frame.
Settings	Input mode, calibration point count, and reading-view (ReaderShell) defaults.

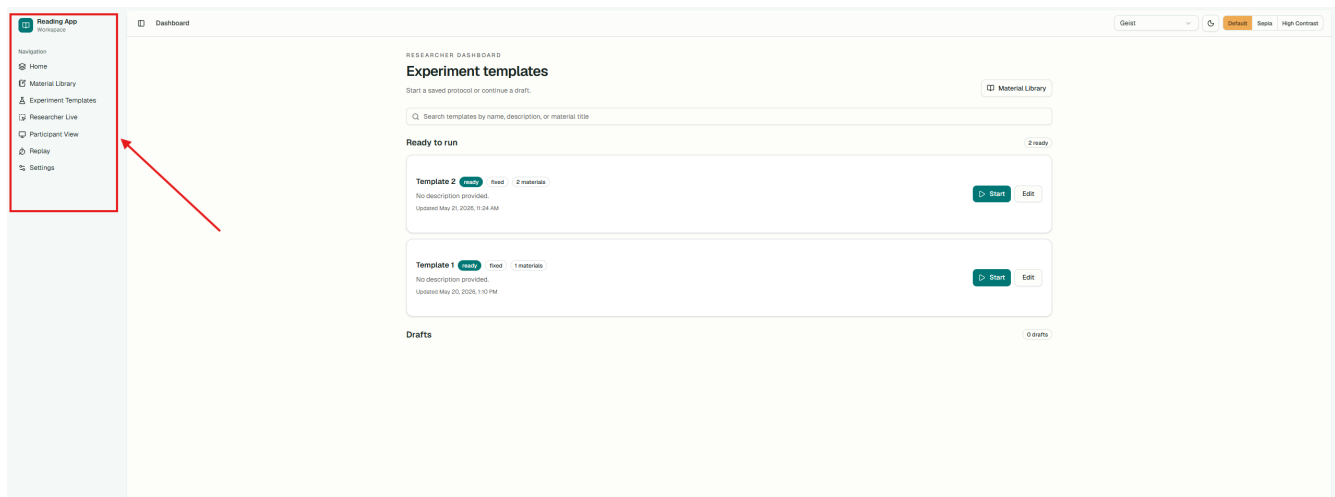


Figure 1: The left sidebar, with every navigation item.

The **top bar** (right side) holds the app font selector, light/dark mode toggle, and color palette toggle. These affect the app appearance.

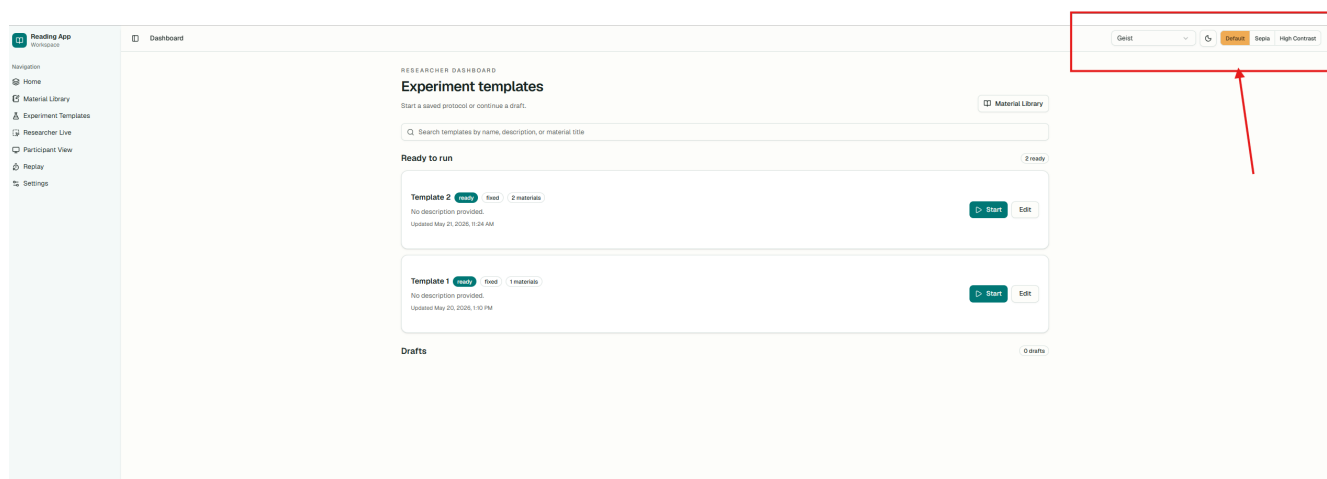


Figure 2: The top bar controls (right side): app font selector, light/dark mode toggle, and color palette.

4. Quick Start (5-minute run)

If you just want to get a session going, follow this condensed path. Each step links to its full section below.

1. **Create at least one reading material** → [§5.1](#)
2. **Build a template** from that material and mark it **Ready** → [§5.2](#)
3. From **Home**, click **Start** on the ready template → [§5.3](#)
4. Open the **Participant View** on the participant's screen → [§5.4](#)
5. Complete **Step 1–4** of the stepper (tracker → baseline → participant info → calibration)
6. Click **Start reading session**, then open **Researcher Live** to monitor → [§5.6](#)
7. When done, **Finish experiment** and **export** the data → [§5.7](#)

5. Step-by-Step Workflows

5.1 Create a Reading Material

A *reading material* is one Markdown text plus its comprehension quiz and presentation defaults, saved as a reusable baseline.

1. In the sidebar, open **Material Library**.
2. Click **New material** (top-right).
3. Fill in the **Reading text** card:
 - **Setup name** — the internal name shown to researchers (required).
 - **Text title** — the title the participant sees (required).
 - **Markdown text** — paste your Markdown, or click **Import .md** to load a file (.md, .markdown, .txt). The title can auto-fill from the filename.
4. (Optional) Add a **Comprehension quiz**. Each question needs at least two options and one marked correct. The quiz is shown to the participant after they finish this text.
5. Adjust **Presentation settings** — font family, font size, line width, line height, letter spacing. Use the **Live preview** panel on the right to see the participant's reading view update as you change values.
6. Toggle **Allow live researcher adjustments**:
 - **On** → a *live-adjustable baseline* (the researcher can still tune typography mid-session).

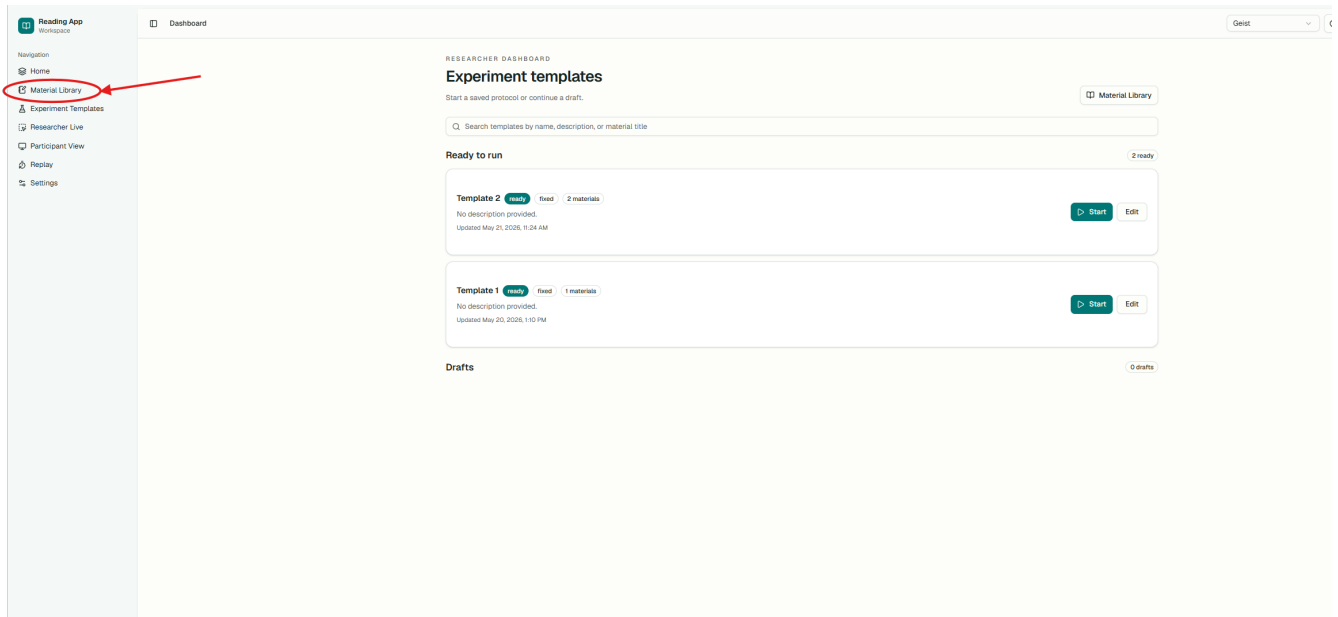


Figure 3: Opening the Material Library from the sidebar.

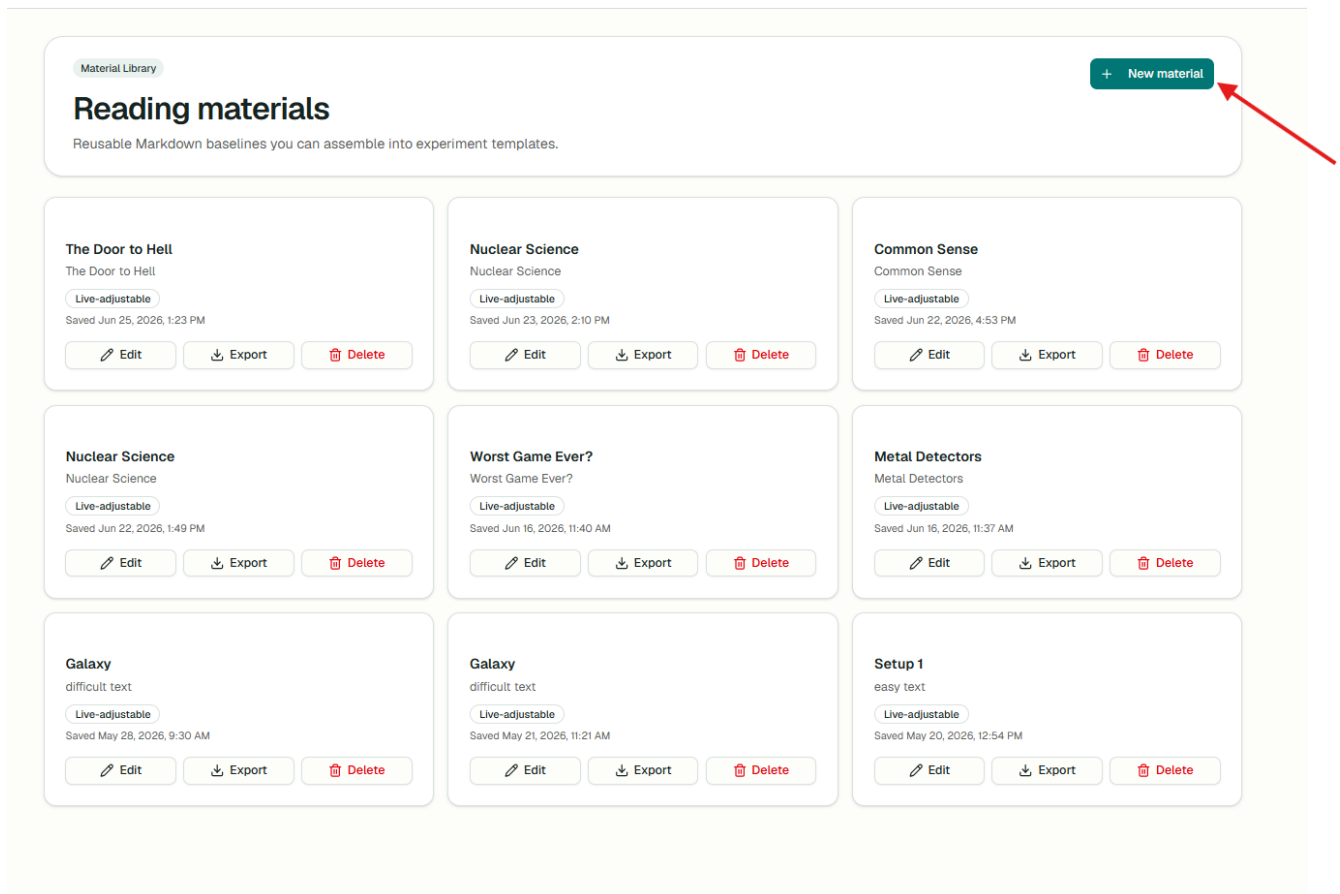


Figure 4: The Material Library listing saved reading materials, with New material at the top right.

Reading text

Edit a local draft here.

Setup name

The Door to Hell

Text title

The Door to Hell

Markdown text

Import .md

The Burning Crater of Turkmenistan

Introduction

The Darvaza Gas Crater, better known as The Door to Hell, is one of the most unusual places on Earth. It is located in the remote Karakum Desert of Turkmenistan. This giant crater has been burning continuously for more than fifty years. At night, the bright orange flames light up the surrounding desert and create a dramatic view. Because of this, the crater attracts travelers, photographers, and scientists from around the world.

How It Started

In 1971, a team of Soviet geologists was exploring the area for natural gas. During drilling operations, the ground suddenly collapsed beneath their equipment. This created a large crater and exposed an underground pocket of methane gas. Methane is flammable and can be dangerous if it spreads into the air. To stop the gas from spreading, the scientists decided to set it on fire. They thought the fire would burn out after only a few days. However, there was much more gas underground than they expected, so the crater is still burning today.

Physical Characteristics

The crater is about 70 meters wide and around 30 meters deep. Many small fires burn inside it at the same time, making it look like a giant furnace. The heat near the crater is very strong, so visitors must be careful and keep a safe distance from the edge.

Tourism

Even though the crater is far away from large cities, it has become one of Turkmenistan's most famous tourist attractions. Many people visit in the evening because the flames are most beautiful after sunset. Some tourists also camp nearby. The desert sky is usually clear, which makes the area a good place for stargazing and night photography.

Environmental Impact

The crater also raises environmental questions. Burning methane turns it into carbon dioxide, but methane is a much stronger greenhouse gas than carbon dioxide. Some experts think burning the gas may be better than allowing methane to escape directly into the atmosphere. Others believe the crater should be closed because it still releases pollution. The government of Turkmenistan has discussed putting out the fire, but it is not an easy task.

Conclusion

The Door to Hell is a fascinating mix of science, history, geology, and human error. What started

Figure 5: The Reading text card with setup name, title, and Markdown pasted in.

Comprehension quiz

Questions shown to the reader after they finish this material. Each question must have at least two options and one marked correct.

Question 1



Prompt

Where is the Door to Hell located?

Options — select the correct answer

- ☐ In the Sahara Desert 
- ☒ In the Karakum Desert of Turkmenistan 
- ☐ In the Amazon rainforest 
- ☐ In northern Canada 

+ Add option

Question 2



Prompt

What caused the crater to form?

Options — select the correct answer

- ☐ A volcanic eruption 
- ☐ A meteor crash 
- ☒ The ground collapsed during drilling 
- ☐ An earthquake split the ground open 

+ Add option

Figure 6: The comprehension quiz editor.

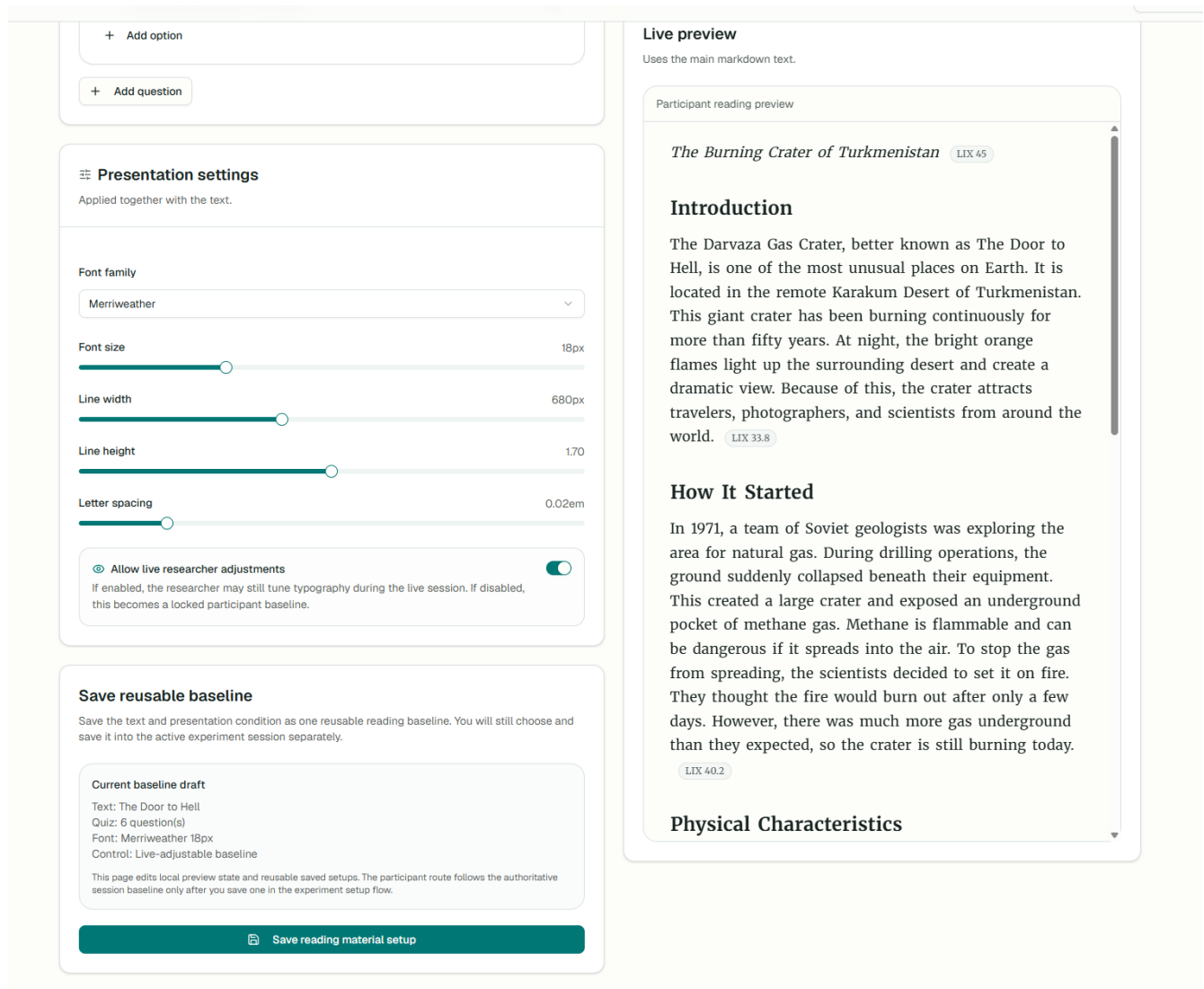


Figure 7: Presentation settings on the left with the live participant preview on the right.

- **Off** → a *locked baseline*.

7. Click **Save reading material setup**. You return to the Material Library.

Import / Export: Use **Import JSON** (top-right of the editor) to load a previously exported material, and the **Export** button on a library card to download one as *.reading-material.json.

5.2 Build an Experiment Template

An *experiment template* is a reusable sequence of texts plus presentation defaults, order mode, runtime strategy, and a calibration requirement.

1. Open **Experiment Templates** → click **New template** (or **Home** → **continue a draft**).

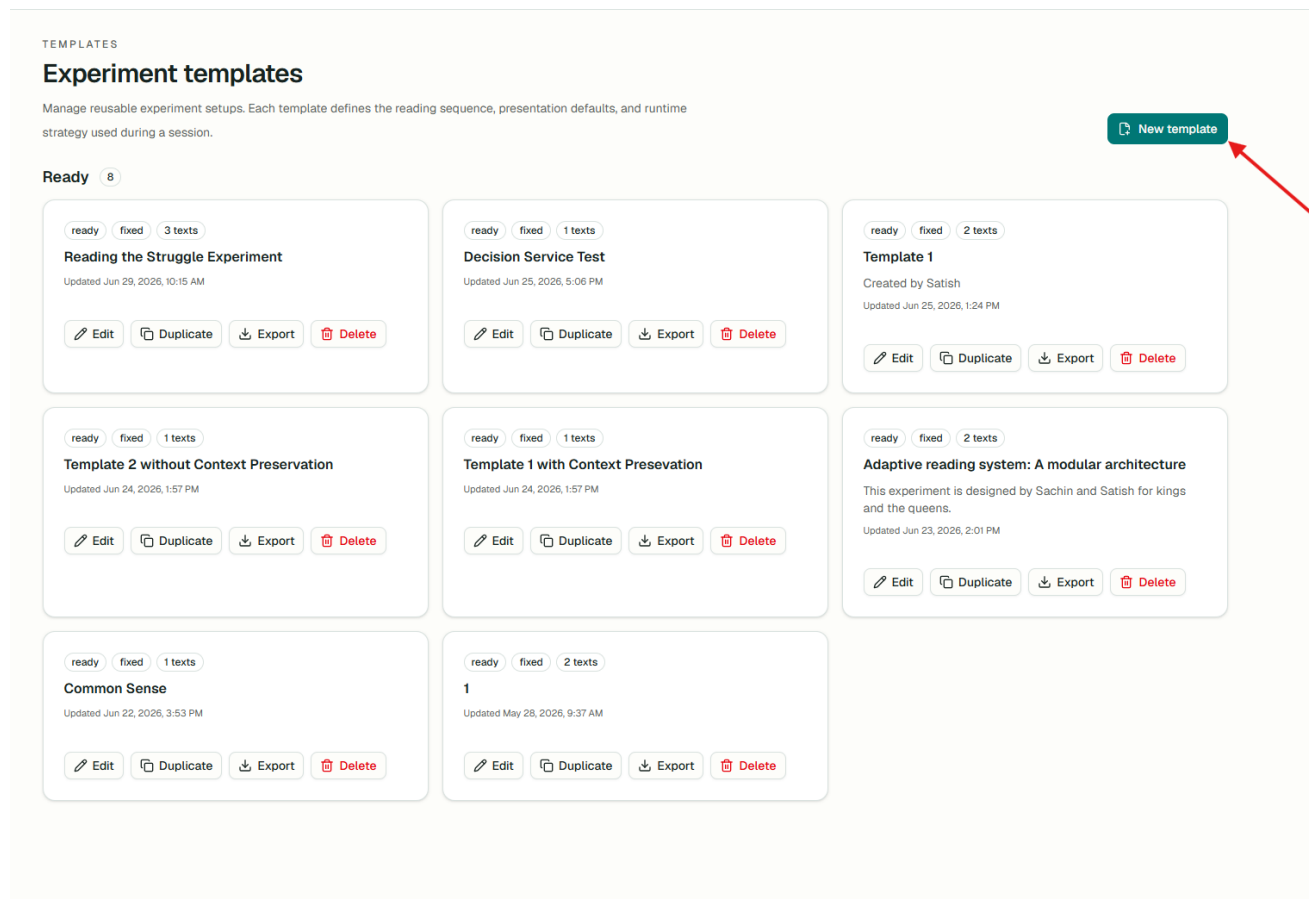


Figure 8: The Experiment Templates library (ready, drafts, archived).

2. In **Template settings**, set:

- **Template name** (required) and an optional **Description**.
- **Status** — Draft, Ready, or Archived. Only **Ready** templates can be started from Home.
- **Material order** — Fixed order or Fully random at session start.
- **Decision strategy** — the runtime decision plugin/mode (see §8).
- **Require calibration** — keep enabled for real eye-tracker sessions.
- **Default font / size / line width / live adjustments** — applied as defaults to added texts.

← Experiment templates

New template

Compose reusable or one-off experiment setups from saved materials. Each template stores copied markdown, resolved presentation settings, order mode, and runtime strategy.

Import JSON

Template settings

Define overview, order, defaults, runtime strategy, and the materials that belong to it.

Template name

Required. This is the reusable setup name shown to researchers.

Example: Danish reading baseline study

Description

Optional notes for researchers about the sequence or purpose of this experiment.

Status

Draft

Material order

Fixed order

Decision strategy

Manual researcher control

Require calibration

Keep enabled for real eye-tracker sessions.

Default font family

Merriweather

Default live adjustments

Allow researchers to adjust baseline display at runtime.

Default font size

18px

Default line width

680px

Add reading material

Material Library

Each added item snapshots the text plus its experiment-specific styling.

Common Sense

Common Sense

Worst Game Ever?

Worst Game Ever?

Metal Detectors

Metal Detectors

Galaxy

difficult text

Galaxy

difficult text

Setup 1

easy text

Experiment sequence

Order matters. The first text becomes the initial live reading baseline.

0 texts

Add one or more saved reading materials to start composing the experiment.

Save as template

Start

Save template

Figure 9: The New template editor: template settings, defaults, and the reading-material picker.

9

3. Under **Add reading material**, click any saved material to add it to the sequence. Each added item *snapshots* the text and its styling into the template.

Figure 10: The reading-material picker grid for adding texts to the template.

4. Arrange the **Experiment sequence**:

- The **first text becomes the initial live reading baseline**.
- Use the up/down arrows to reorder, the trash icon to remove.
- Per text, override **displayed title, font, size, line width, line height, letter spacing**, and **Allow live presentation adjustments**.

5. Finish:

- **Save template** (or **Update template**) — keeps you in the library.
- **Start** — saves and jumps straight into a session for this template. (Turn off **Save as template** first if you want a one-off run that is not stored.)

Duplicate / Export: From the template library, use **Duplicate** to clone a setup as a new draft, or **Export** to download it as *.experiment-template.json. **Import JSON** is at the top-right of the template editor.

Experiment sequence

Order matters. The first text becomes the initial live reading baseline.

2 texts

Text 1

The Door to Hell

Quiz: 6 questions

↑

↓

🗑️

The Door to Hell

Source material: The Door to Hell

Displayed title

The Door to Hell

Font family

Merriweather

Font size

18px

Line width

680px

Line height

1.70

Letter spacing

0.02em

Allow live presentation adjustments

If turned off, this text starts from a locked baseline during the live session.

☒

Text 2

Worst Game Ever?

Quiz: 3 questions

↑

↓

🗑️

Worst Game Ever?

Source material: Worst Game Ever?

Displayed title

Worst Game Ever?

Font family

Merriweather

Font size

18px

Line width

680px

Line height

1.70

Letter spacing

0.02em

Allow live presentation adjustments

If turned off, this text starts from a locked baseline during the live session.

☒

☒ Save as template

Figure 11: The experiment sequence: ordered text cards with per-text presentation overrides.

5.3 Run an Experiment Session

Sessions are driven by a **4-step stepper**. The researcher completes Steps 1–2 (preparation); the participant completes Steps 3–4. Start a session by clicking **Start** on a ready template (Home), or **Start** from the template editor.

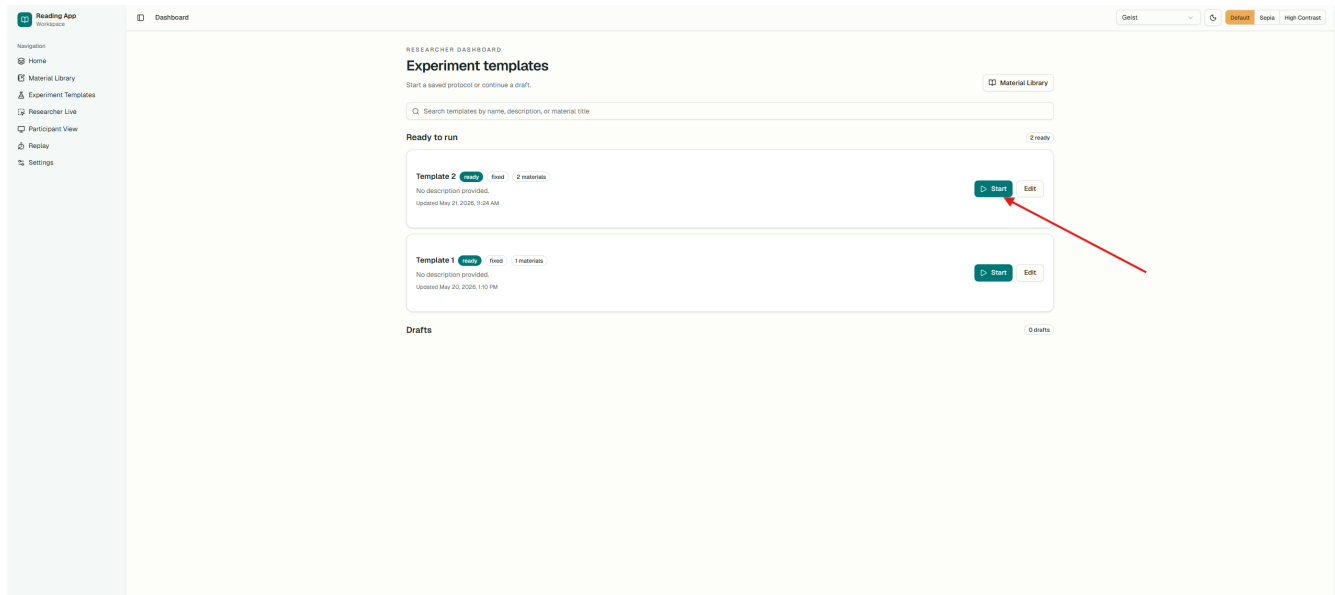


Figure 12: Start a session by clicking Start on a ready template from the dashboard.

The left rail shows all four steps with status badges (Done, Current, Available, Locked) and an owner badge (Researcher / Participant).

Step 1 — Choose eyetracker (*Researcher*)

This step's content depends on the active **input mode** (Settings → Input mode). In **Mouse mode** it shows a "Mouse input is active" card and completes automatically. The steps below describe full **Eyetracker** mode (also used for **Eyetracker + face**).

1. Pick the device from the **eyetracker** dropdown (shown as *name / model / serial*). Click the refresh icon to re-scan if your device is missing.
2. Upload the device **licence** file (drag-and-drop or browse). Optionally tick **save licence** and/or **overwrite existing licence**.
3. Confirm the selection. Click **Save researcher setup** to continue.

Step 2 — Reading material (*Researcher*)

1. Pick a **reusable experiment** (Ready templates appear as cards). Its first text becomes the live reading baseline. (*If you started from a template, this selection is locked.*)
2. Confirm the **Runtime plugins**:
 - **Decision plugin** — Manual / Rule-based (advisory or autonomous) / external Decision-maker.
 - **Eye analyzer** — Built-in analyzer or an external Eye analyzer service.
 - Status badges show whether each external service is *Connected*, *Built-in*, or *Unavailable*.
3. Review the selection summary, then click **Apply baseline & continue**.

Researcher experiment control

Prepare the device and reading baseline first, then follow the participant steps and start the session when everything is ready.

EXPERIMENT FLOW

Researcher preparation first, participant setup second.



Choose eyetracker

Researcher Current

STEP 1

Select the device and its licence so the session can talk to the tracker.

Select and license an eye tracker before starting the session.



Reading material

Locked

STEP 2

Choose the text, apply the baseline, and confirm runtime plugins.

Complete the eyetracker setup before preparing the reading baseline.



Participant info

Locked

STEP 3

Record the participant details after researcher preparation is complete.

Waiting for the researcher to finish preparation first.



Calibration

Locked

STEP 4

Open the calibration page, complete it, and come back here when it is done.

Waiting for participant

Step 1

Eyetracker

Choose and prepare the eyetracker.

Select the eyetracker used for this session and upload the associated license when needed.

ⓘ Eyetracker step still needs attention

Select and license an eye tracker before starting the session.

Select an eyetracker

Pick the device model used in this experiment.

Eyetracker

Reload

/ IS4_Large_Peripheral / IS404-100106341184

License status

Upload a valid license file for / IS4_Large_Peripheral / IS404-100106341184.

ⓘ Existing license detected

The system already has a saved license for this eyetracker. To prevent accidental replacement, uploading is locked until you explicitly enable overwrite.

☐ I want to overwrite the existing license

Enable this only if you intentionally want to replace the saved license.

License file



Upload is locked

Enable overwrite above to upload a replacement license.

☐ Save this license for future use

Next time you select this eyetracker, you will not need to upload the license again.

Previous

Save researcher setup

Figure 13: Step 1: choosing and preparing the eyetracker, including licence upload.

Researcher experiment control

Prepare the device and reading baseline first, then follow the participant steps and start the session when everything is ready.

EXPERIMENT FLOW

**Researcher preparation first,
participant setup second.**



Choose eyetracker

Researcher Done

STEP 1

Select the device and its licence so the session can talk to the tracker.

Using selected eyetracker with licence ready.



Reading material

Current Researcher

STEP 2

Choose the text, apply the baseline, and confirm runtime plugins.

Choose the reading material before starting the session.



Participant info

Locked Participant

STEP 3

Record the participant details after researcher preparation is complete.

Waiting for the researcher to finish preparation first.



Calibration

Locked Participant

STEP 4

Open the calibration page, complete it, and come back here when it is done.

Waiting for participant information before calibration can begin.

Step 2 Researcher Experiment baseline

Choose a reusable experiment and runtime plugins.

Pick the reusable experiment, apply its first text as the live reading baseline, and confirm the runtime plugins before handing over to the participant.

Reading material locked

This session was started from a template, so its reading baseline stays fixed here.

Reading the Struggle Experiment

3 texts in sequence

Reusable experiment

Decision Service Test

1 texts in sequence

Reusable experiment

Template 1

2 texts in sequence

Reusable experiment

Template 2 without Context Preservation

1 texts in sequence

Reusable experiment

Template 1 with Context Preservation

1 texts in sequence

Reusable experiment

Adaptive reading system: A modular architecture

2 texts in sequence

Reusable experiment

Common Sense

1 texts in sequence

Reusable experiment

1

2 texts in sequence

Reusable experiment



Create experiment

Runtime plugins

Choose the decision and eye movement analysis plugins for this session.

Decision: Built-in

Analyzer: Built-in

Decision plugin

No decision plugin

Eye analyzer

Built-in plugin

Manual control

Built-in analyzer

External decision plugins become selectable when a service connects.

The built-in analyzer remains active until an external analyzer is selected.

Selected: Worst Game Ever? | Reading the Struggle Experiment | Manual control | Reusable experiment | Live-adjustable | Reading the Struggle Experiment (3 texts)

Previous

Apply baseline & continue

Figure 14: Step 2: selecting a reusable experiment and confirming the runtime plugins.

Step 3 — Participant info (*Participant*) Recorded on the participant's screen (or by the researcher). Fields: **Name, Age, Sex, Existing eye condition, Reading proficiency**. All are required and validated. Click **Continue to calibration**.

Researcher experiment control

Prepare the device and reading baseline first, then follow the participant steps and start the session when everything is ready.

EXPERIMENT FLOW

Researcher preparation first, participant setup second.

Choose eyetracker Researcher Done
STEP 1
Select the device and its licence so the session can talk to the tracker.
Using selected eyetracker with licence ready.

Reading material Done Researcher
STEP 2
Choose the text, apply the baseline, and confirm runtime plugins.
Worst Game Ever? is saved for session start.

Participant info Current Participant
STEP 3
Record the participant details after researcher preparation is complete.
Save the participant information before starting the session.

Calibration Locked Participant
STEP 4
Open the calibration page, complete it, and come back here when it is done.
Waiting for participant information before calibration can begin.

Capture participant information.
Record the participant details.

Name
Satish

Age
18

Sex
Male

Existing eye condition
None

Reading proficiency
Intermediate

Choose the participant's reading level for baseline grouping.

Previous Continue to calibration

Figure 15: Step 3: the participant information form.

Step 4 — Calibration (*Participant*) Opens the full-screen calibration routine. See §5.5. When validation passes, the participant returns here automatically.

Start the session Once all four steps show **Done**, the final button becomes **Start reading session**. Clicking it: - (Researcher start) → opens **Researcher Live**. - (Participant start) → opens the **Reading** page on the participant screen.

5.4 The Participant View

Open **Participant View** from the sidebar (it opens in a new tab) and place it on the participant's screen.

Researcher experiment control

Prepare the device and reading baseline first, then follow the participant steps and start the session when everything is ready.

EXPERIMENT FLOW

Researcher preparation first, participant setup second.

Choose eyetracker

ResearcherDone

STEP 1

Select the device and its licence so the session can talk to the tracker.

Using selected eyetracker with licence ready.

Reading material

Researcher

DoneSTEP 2

Choose the text, apply the baseline, and confirm runtime plugins.

difficult text is saved for session start.

Participant info

Participant

DoneSTEP 3

Record the participant details after researcher preparation is complete.

Participant Satish saved to the session.

Calibration

Participant

DoneSTEP 4

Open the calibration page, complete it, and come back here when it is done.

Validation passed with No rating yet.

Step 4ParticipantCalibrationRead only

Run the Tobii calibration and validation.

Calibration can only be completed from the participant view.

Participant-owned step

The participant completes calibration from the participant view. Status updates will appear here automatically.

Calibration applied Validation passed

The selected eye tracker has an active validated calibration with no rating yet and 0 validation samples.

AVERAGE ACCURACY

Not available

AVERAGE PRECISION

Not available

COMPLETED

Not available

VALIDATION QUALITY

No rating yet

Previous

Start reading session

Figure 16: All four steps complete, with the reading session ready to start.

16

- Before the researcher begins setup, the page shows a **waiting** card (“Experiment has not started yet” / “Your session is being prepared”).

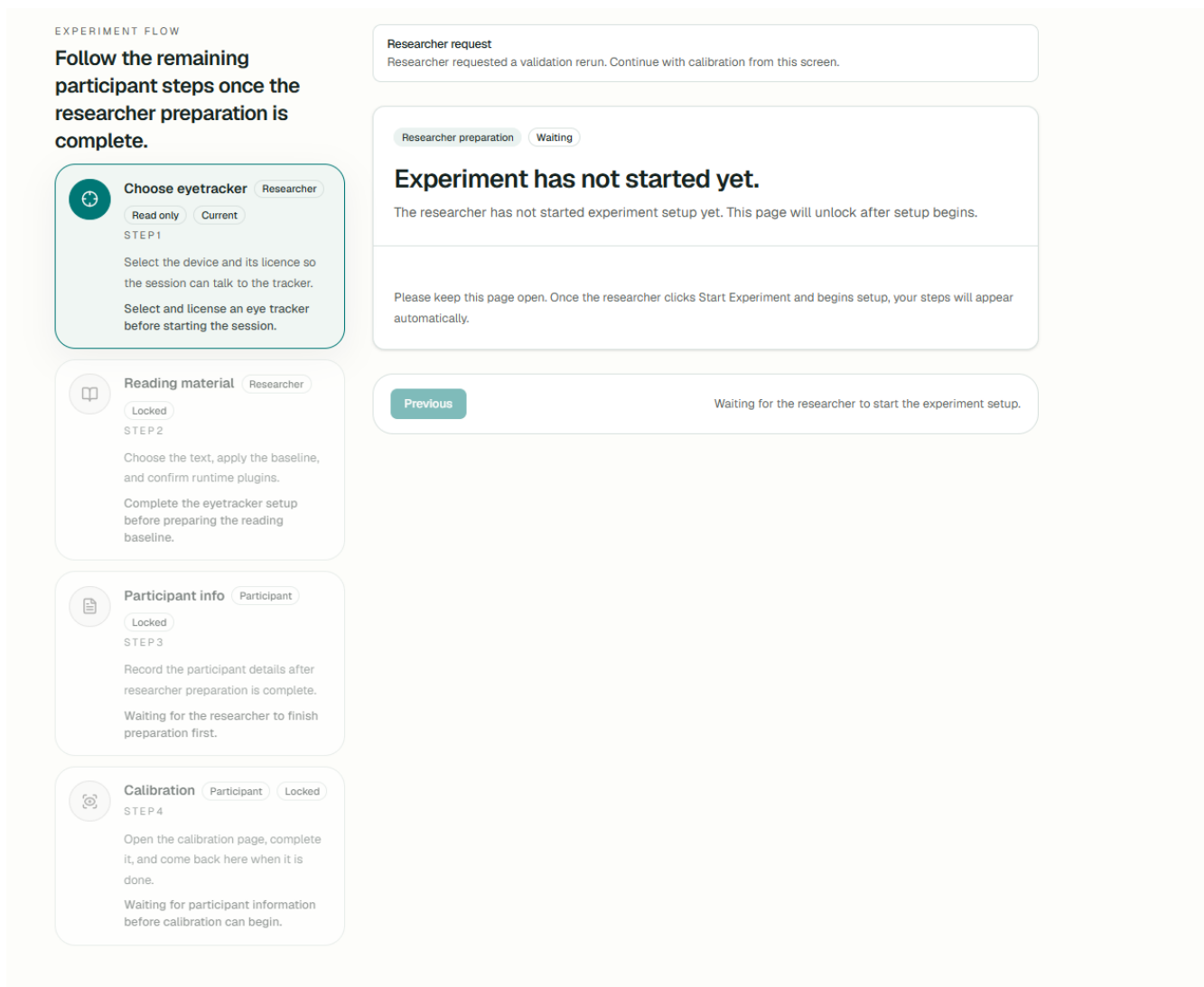


Figure 17: The participant view before the researcher has started setup (“Experiment has not started yet”).

- Once researcher preparation (Steps 1–2) is ready, the participant’s steps (3–4) unlock automatically.
- After participant info and calibration are complete, the page shows “**Waiting for the researcher to start the session.**”

The reading page begins automatically when the researcher starts the session.

Re-run calibration: If the researcher requests a calibration or validation re-run, the participant sees a notice and is routed back into calibration.

5.5 Calibration

Calibration maps the participant’s gaze to the screen and then **validates** the result against a quality threshold. It runs in **full screen** and must stay visible — leaving full screen or hiding the tab interrupts the run.

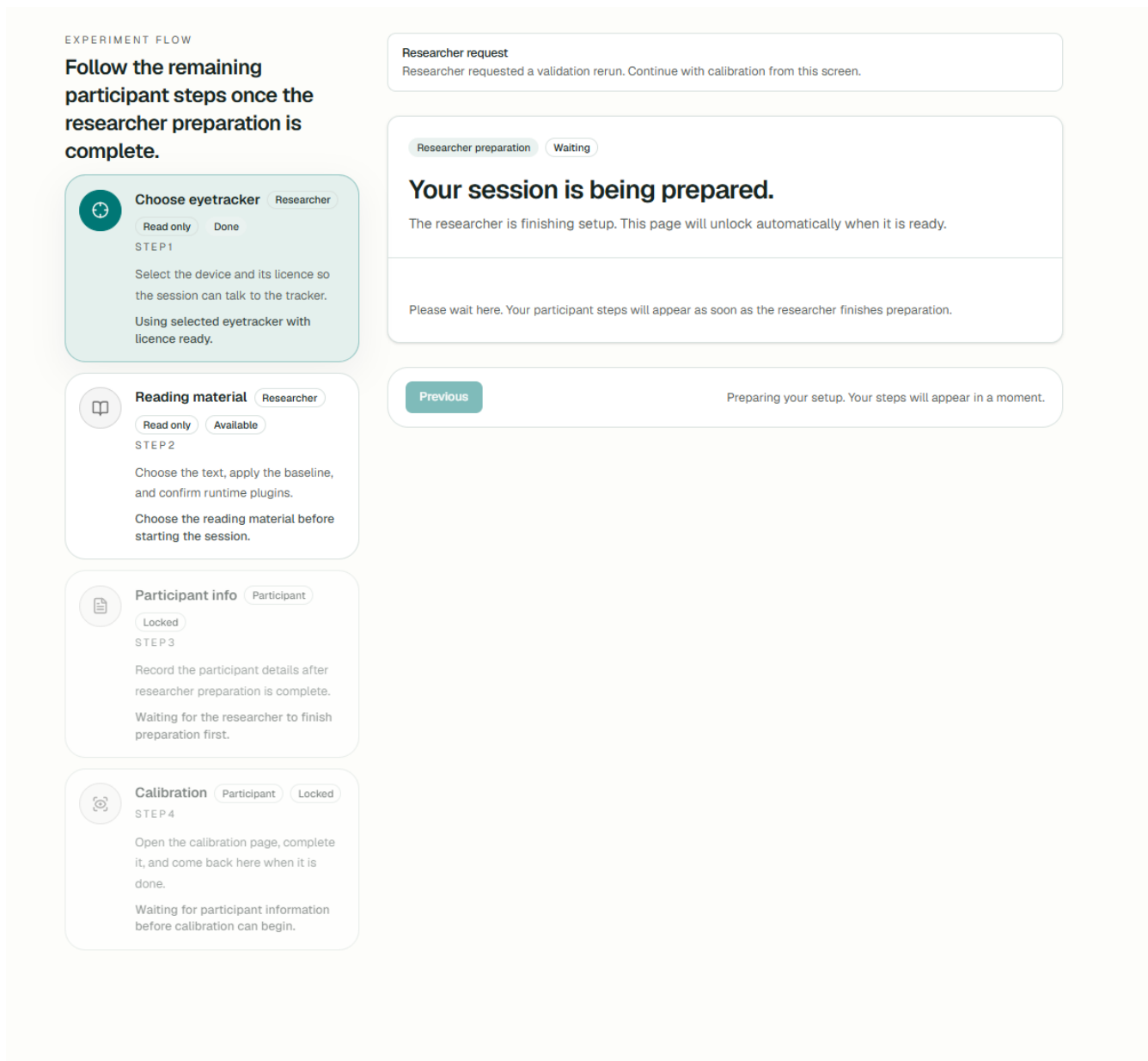


Figure 18: Once the researcher begins setup, the participant sees “Your session is being prepared”.

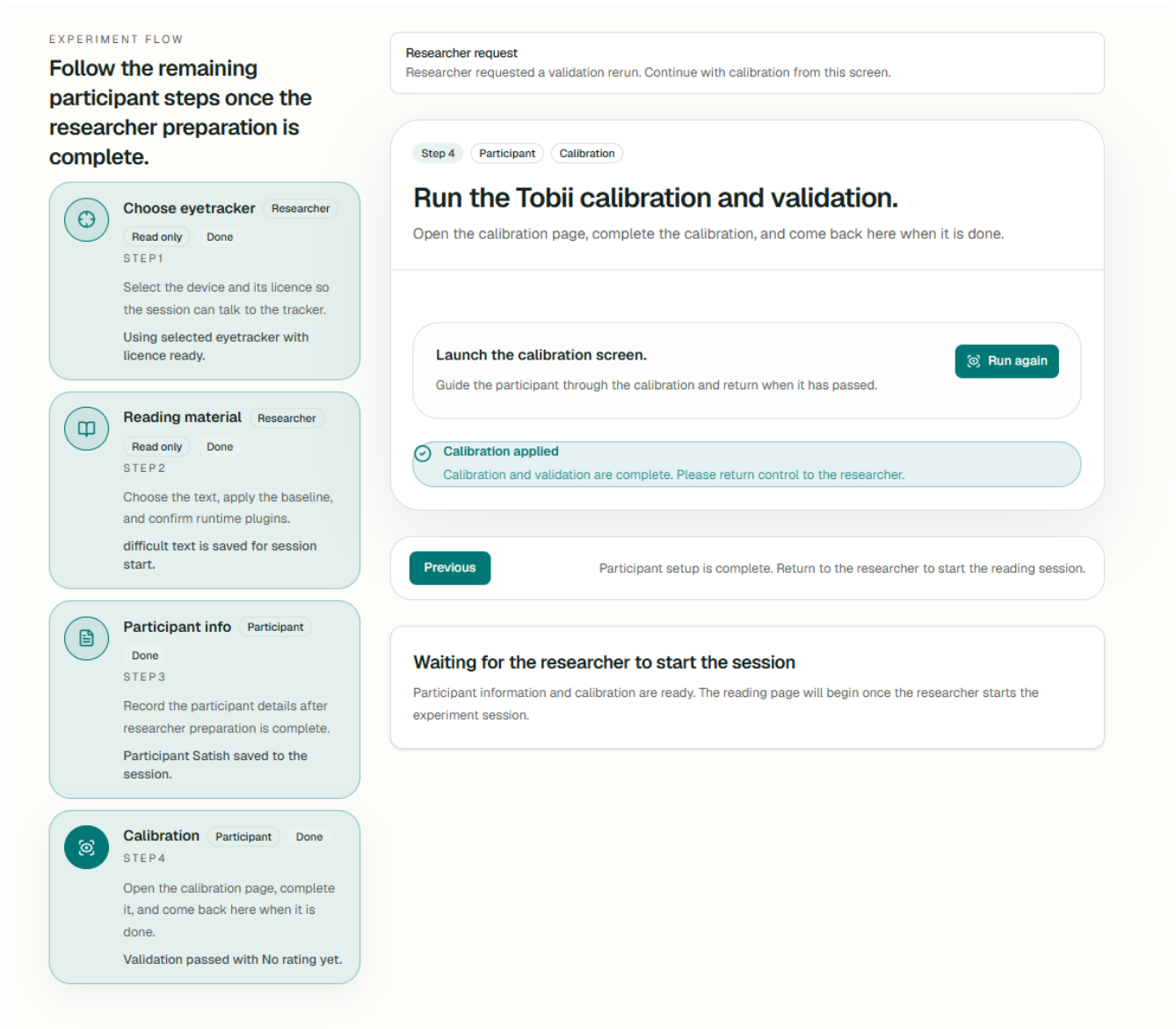


Figure 19: The participant waiting screen shown once setup is complete, before the session starts.

1. On the **Ready** screen, the participant looks at the center of each target. A gaze-preview overlay helps confirm tracking. Click **Start**.

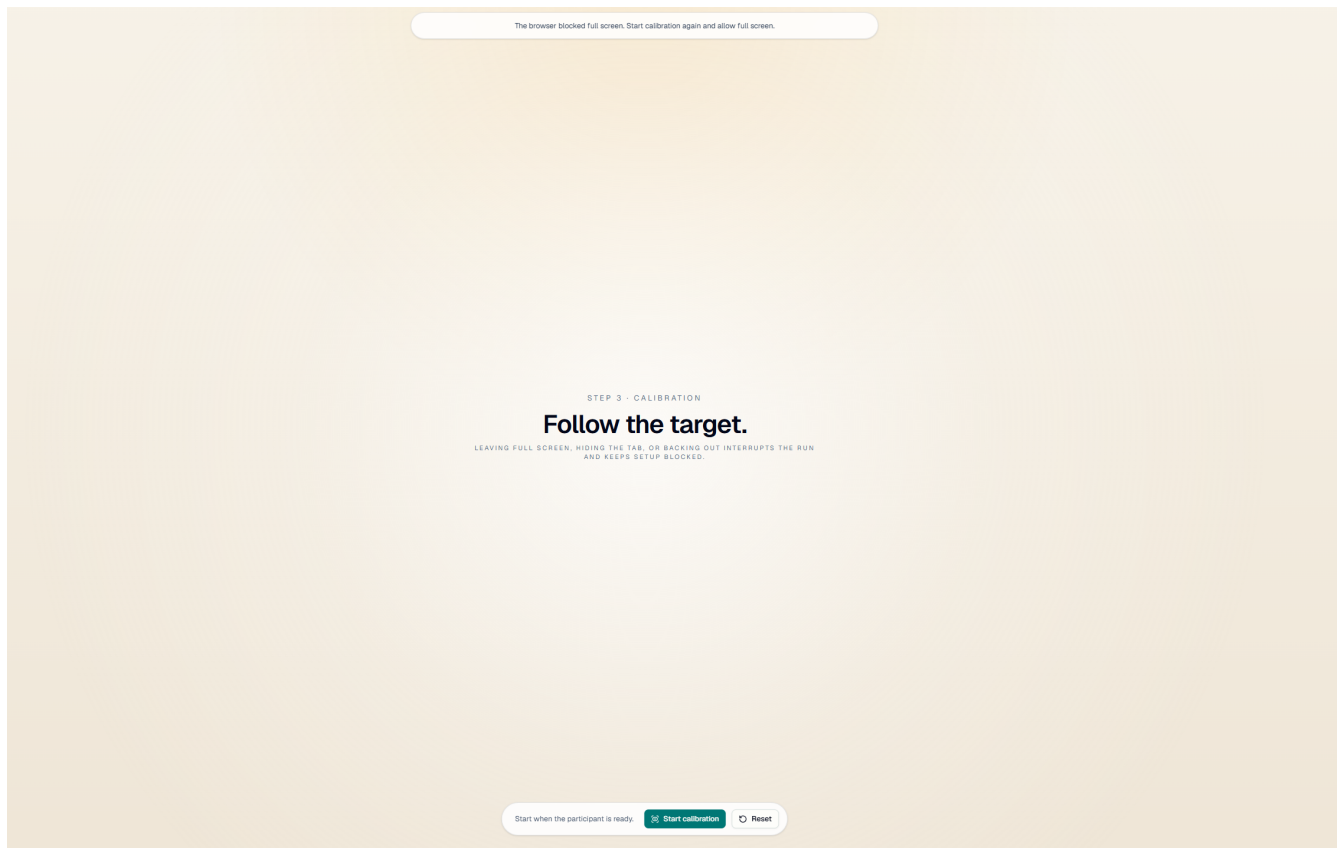


Figure 20: The calibration ready screen with the gaze-preview overlay.

2. **Calibration run** — a target moves point-to-point; the participant holds their gaze on each.
3. **Validation run** — repeats with validation targets to measure **accuracy** and **precision**.
4. **Review** (*researcher mode*) — shows quality, average accuracy/precision, and sample count.

- If it **passed**, accept and return to the workflow.
- If it **failed**, **Rerun validation** or **Start** a fresh calibration.

5. If a run is interrupted or rejected, a **failure** panel offers **Reset** or **Return to setup**.

Note: In participant mode, a passing validation routes the participant straight back to their setup; the metrics review is a researcher-mode step.

5.6 The Live Researcher View

Open **Researcher Live** to monitor the active session. It runs full screen and is laid out in three columns.

Left column — Live controls & signals - Participant name and connection status. - **Sample rate (Hz)**, **gaze validity rate**, and **latency (ms)**. - **Struggle signals** (when the external eye analyzer is connected and selected).

Center column — Live reader (mirror) - A real-time mirror of what the participant sees, with their gaze focus highlighted. - **Follow participant** keeps the mirror locked to the participant's scroll position. - A

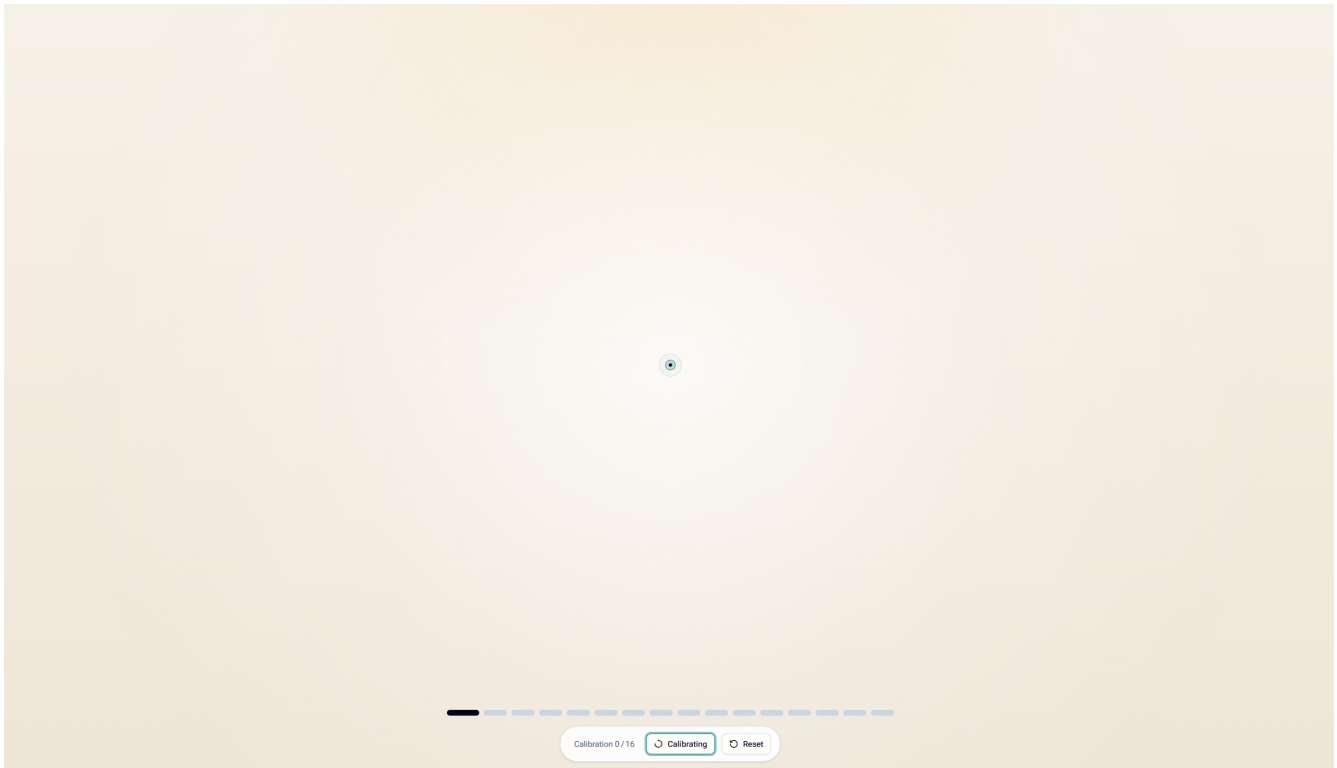


Figure 21: A calibration run in progress: the participant follows the moving target while the progress dots track collected points.

trust banner appears when the mirror is only *approximate* (e.g. not full screen) — click **Enter full screen** to restore an exact mirror. - Optional overlays (fixation heatmap, saccade path / “reading dynamics”) via the reader controls.

Right column — Interventions - **Trigger an intervention** manually from the available intervention modules (e.g. typography or appearance changes), with a reason. - **Approve / reject** decision proposals coming from a decision plugin (advisory mode). - **Apply pending intervention now** to commit a queued change immediately. - Set the **intervention commit boundary** (when a layout change is allowed to take effect). - **Advance to the next text** in a multi-text experiment sequence.

5.7 Finishing & Exporting a Session

When the session is finished:

1. In the **Researcher Live** view (or its empty/complete state), click **Finish experiment**.
2. After finishing, the completion actions appear. You can:
 - **Start new experiment** — resets state and returns to the setup flow.
 - **Download** the data as **JSON**, **CSV**, **Processed**, or **Telemetry**.
 - **Save** the replay export (JSON or CSV) into the app under a name, so it appears in **Replay**.

Naming: The export name defaults to the reading title or participant name; edit it before saving or downloading.

Researcher experiment control

Prepare the device and reading baseline first, then follow the participant steps and start the session when everything is ready.

EXPERIMENT FLOW

Researcher preparation first,
participant setup second.



Choose eyetracker

Researcher Done

STEP 1

Select the device and its licence so the session can talk to the tracker.

Using selected eyetracker with licence ready.



Reading material

Done Researcher

STEP 2

Choose the text, apply the baseline, and confirm runtime plugins.
difficult text is saved for session start.



Participant info

Done Participant

STEP 3

Record the participant details after researcher preparation is complete.

Participant Satish saved to the session.



Calibration

Done Participant

STEP 4

Open the calibration page, complete it, and come back here when it is done.

Validation passed with Fair quality.

Step 4 Participant Calibration Read only

Run the Tobii calibration and validation.

Calibration can only be completed from the participant view.

Participant-owned step

The participant completes calibration from the participant view. Status updates will appear here automatically.

Calibration applied Validation passed

The selected eye tracker has an active validated calibration with fair quality and 150 validation samples.

AVERAGE ACCURACY

0.52°

AVERAGE PRECISION

0.16°

COMPLETED

6/29/2026, 12:55:01 PM

VALIDATION QUALITY

Fair quality

Point-by-point review

Review each calibration point before deciding if a rerun is needed.

Samples: 150

Upper left

(0.28, 0.24) - 30 samples

Good quality

ACCURACY

0.25°

PRECISION

0.13°

Upper right

(0.72, 0.24) - 30 samples

Good quality

ACCURACY

0.32°

PRECISION

0.16°

Center

(0.50, 0.50) - 30 samples

Fair quality

ACCURACY

0.51°

PRECISION

0.15°

Lower left

(0.28, 0.76) - 30 samples

Fair quality

ACCURACY

0.76°

PRECISION

0.17°

Lower right

(0.72, 0.76) - 30 samples

Fair quality

ACCURACY

0.75°

PRECISION

0.18°

Run validation again

Run full calibration again

Previous

Start reading session

Figure 22: The calibration review panel showing quality, accuracy, and precision metrics.

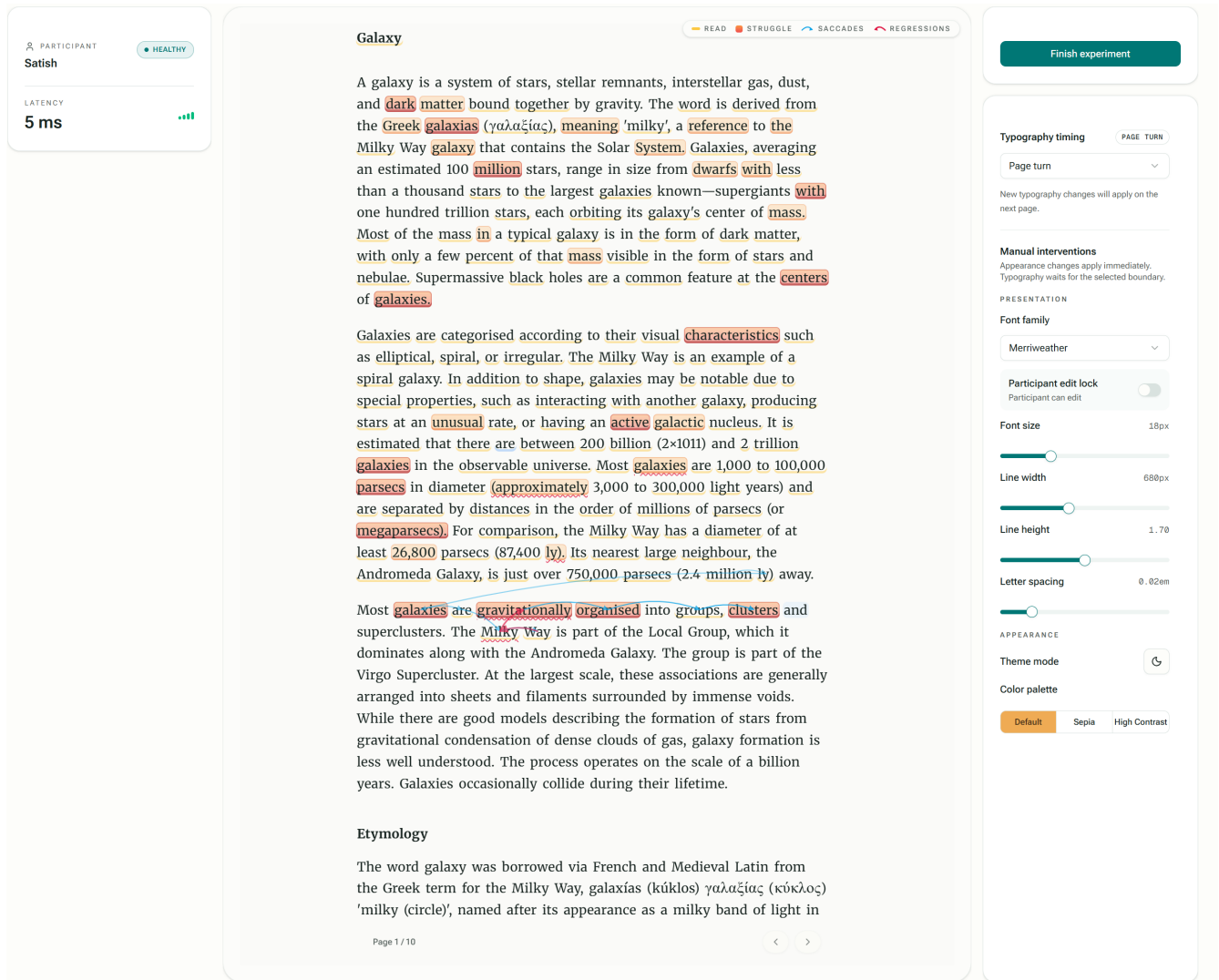


Figure 23: The Researcher Live view: signals on the left, the live participant mirror in the center, and intervention controls on the right.

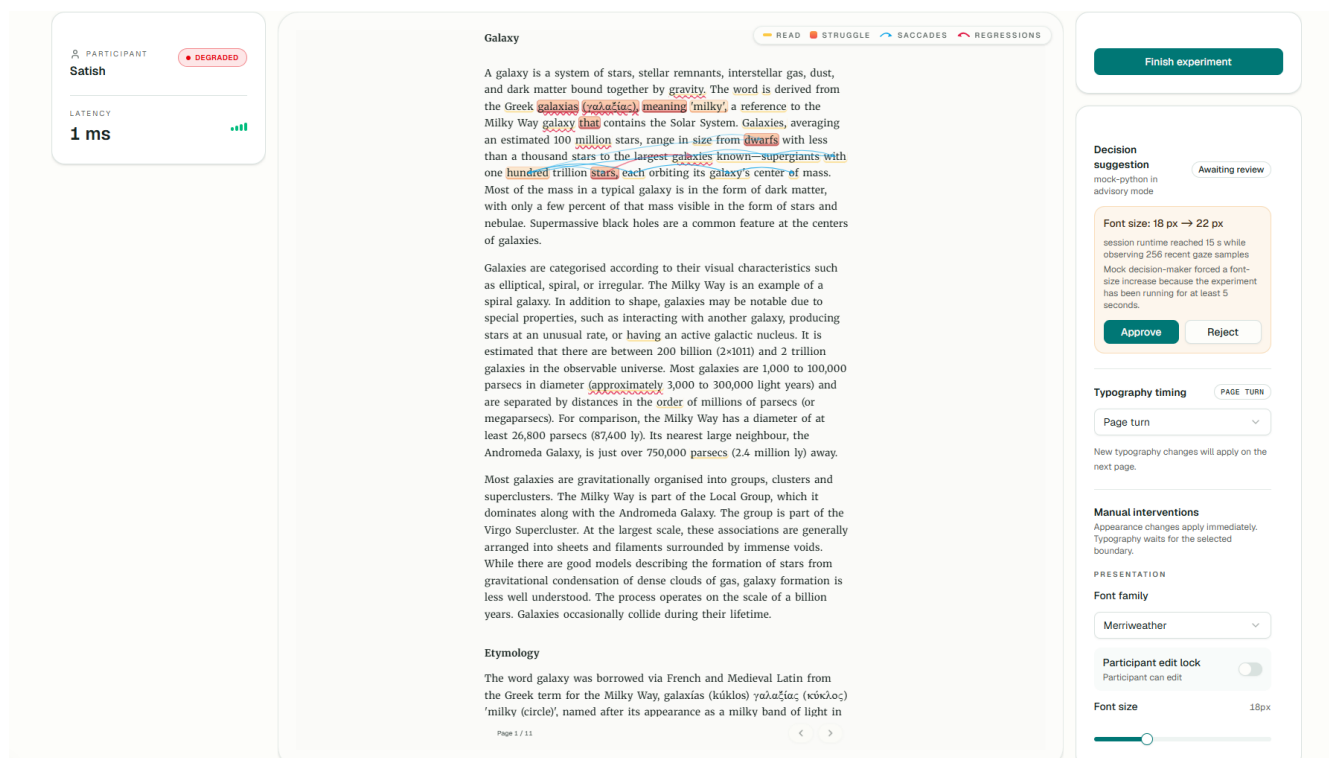


Figure 24: The interventions column with the decision-provider proposal awaiting approval.

5.8 Replay a Saved Session

Open **Replay** to review a recorded session.

1. Load a session:

- Drag-and-drop or browse for an exported file, **or**
- Pick one of the **saved exports** listed on the upload screen.
- You can also **convert** a saved export into a processed report from here.

2. The replay opens in a three-column layout:

- **Left** — playback controls: play/pause, restart, scrubber, **playback speed**, reader overlay options, and load/clear.
- **Center** — the reconstructed reading view at the current time, including gaze focus, saccade paths, and any quiz/finish screens that occurred.
- **Right** — metadata and a **key-events timeline** (click an event to seek), plus oculomotor counts (**fixations, saccades, regressions**).

6. Settings

Open **Settings** from the sidebar. There are three sections (tabs):

Section	What it controls
Input mode	The sensing source: Use eyetracker (Tobii), Use eyetracker + face (Tobii gaze + webcam facial signals), or Use mouse mode (demo). Choose, then Save . See §7.
Calibration	The number of calibration points used during a calibration run.

Section	What it controls
ReaderShell	View-specific reading defaults for the researcher mirror and the replay reader (e.g. which overlays are on by default).

The chosen input mode changes **Step 1** of the experiment stepper accordingly.

7. Sensing Modes Explained

Mode	Badge	Behavior
Use eyetracker	Tobii	Requires tracker selection, licence handling, calibration, and validation. The standard real-experiment mode.
Use eyetracker + face	Hybrid	Tobii stays the authoritative gaze source; a webcam adds facial strain/expression signals. If the webcam is unavailable, gaze still runs but facial signals are degraded.
Use mouse mode	Demo	The participant's mouse position is used as synthetic gaze. Skips tracker selection, licence, and hardware checks — ideal for demos without hardware.

8. Runtime Plugins (Decision & Eye Analysis)

The platform separates **who decides on interventions** from **who analyzes eye movement**. Both are selectable per session in **Step 2**, and a default decision strategy is stored on each template.

Decision plugin options:

Option	Meaning
Manual control	Interventions stay fully researcher-operated. No decision plugin.
Rule-based advisory	Built-in rules <i>propose</i> interventions for the researcher to approve.
Rule-based autonomous	Built-in rules <i>apply</i> supported interventions automatically.
Decision-maker advisory	A connected external service proposes interventions.
Decision-maker autonomous	A connected external service requests interventions automatically.

Eye analyzer options:

Option	Meaning
Built-in analyzer	Backend thresholds provide fixation/saccade state.
Eye analyzer service	A connected external service supplies fixation/saccade and struggle state.

External options are only selectable when the corresponding service is **connected** and supports the chosen mode. Otherwise the option shows — **unavailable** and the built-in option remains active.

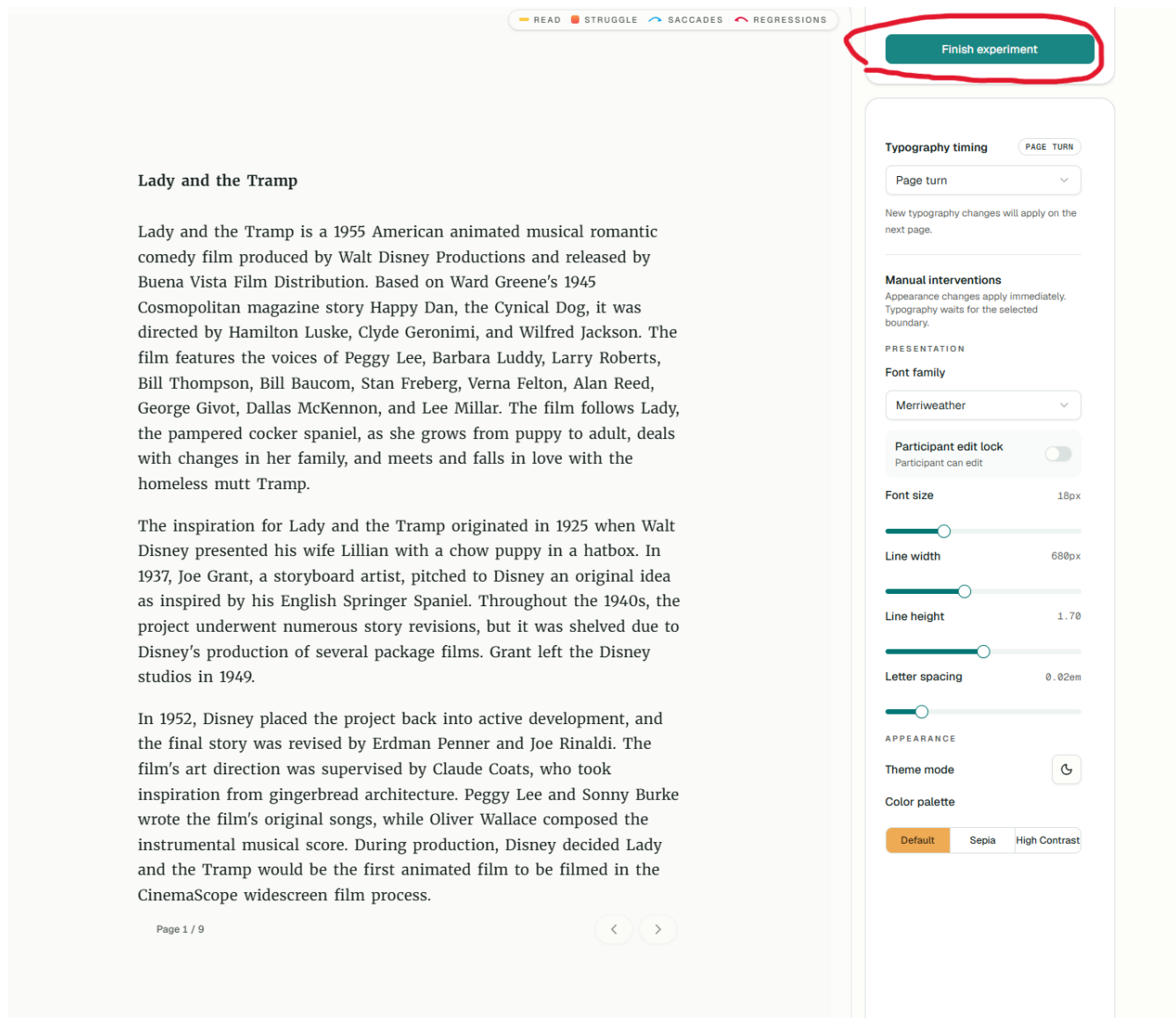


Figure 25: The Finish experiment button in the live view, used to end the session.

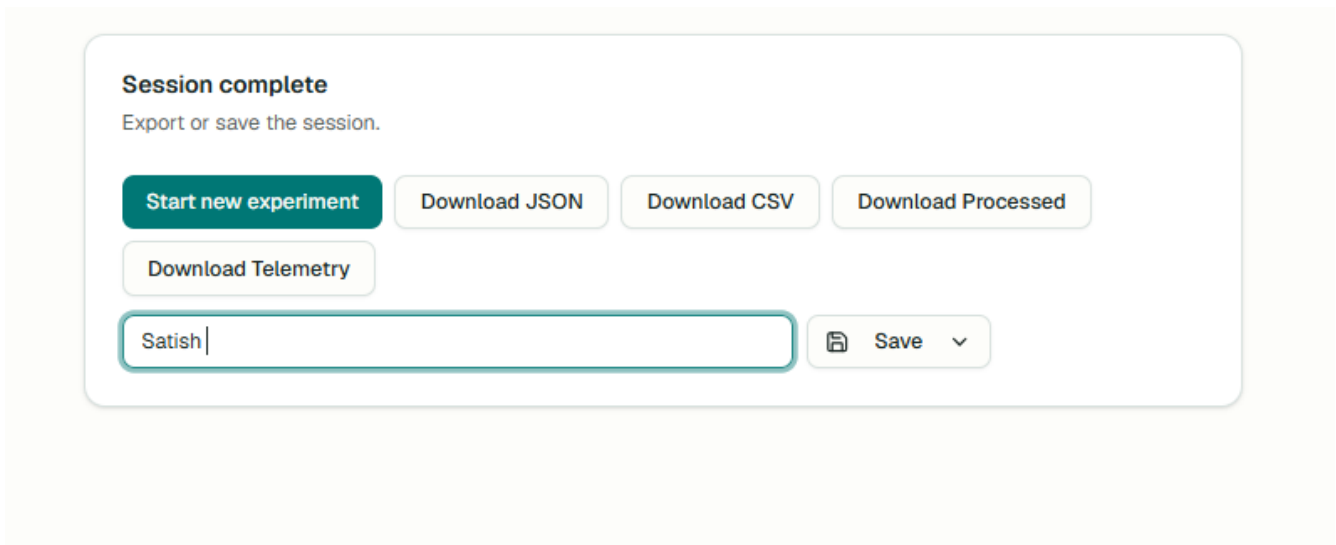


Figure 26: The session-complete card: export/download buttons and the export name + Save field.

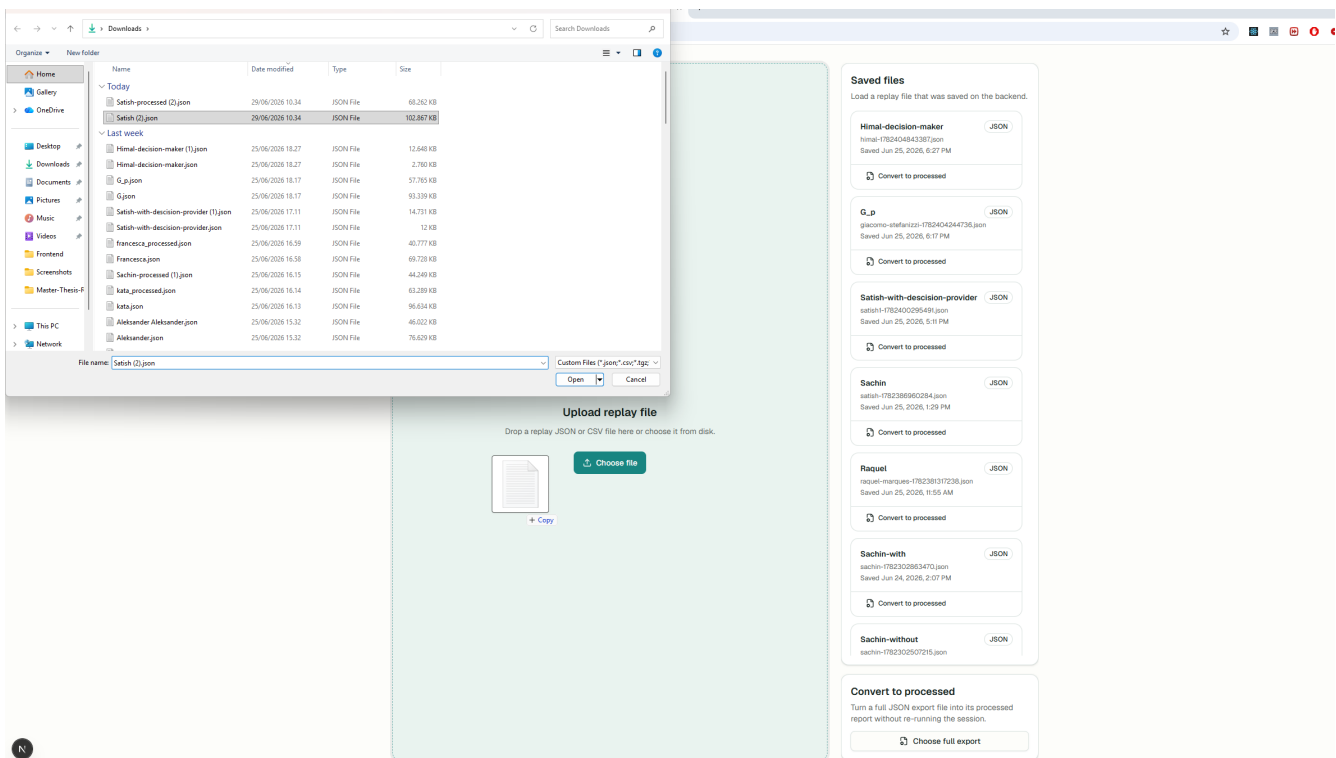


Figure 27: The replay upload screen: drag-and-drop area plus the list of saved exports.

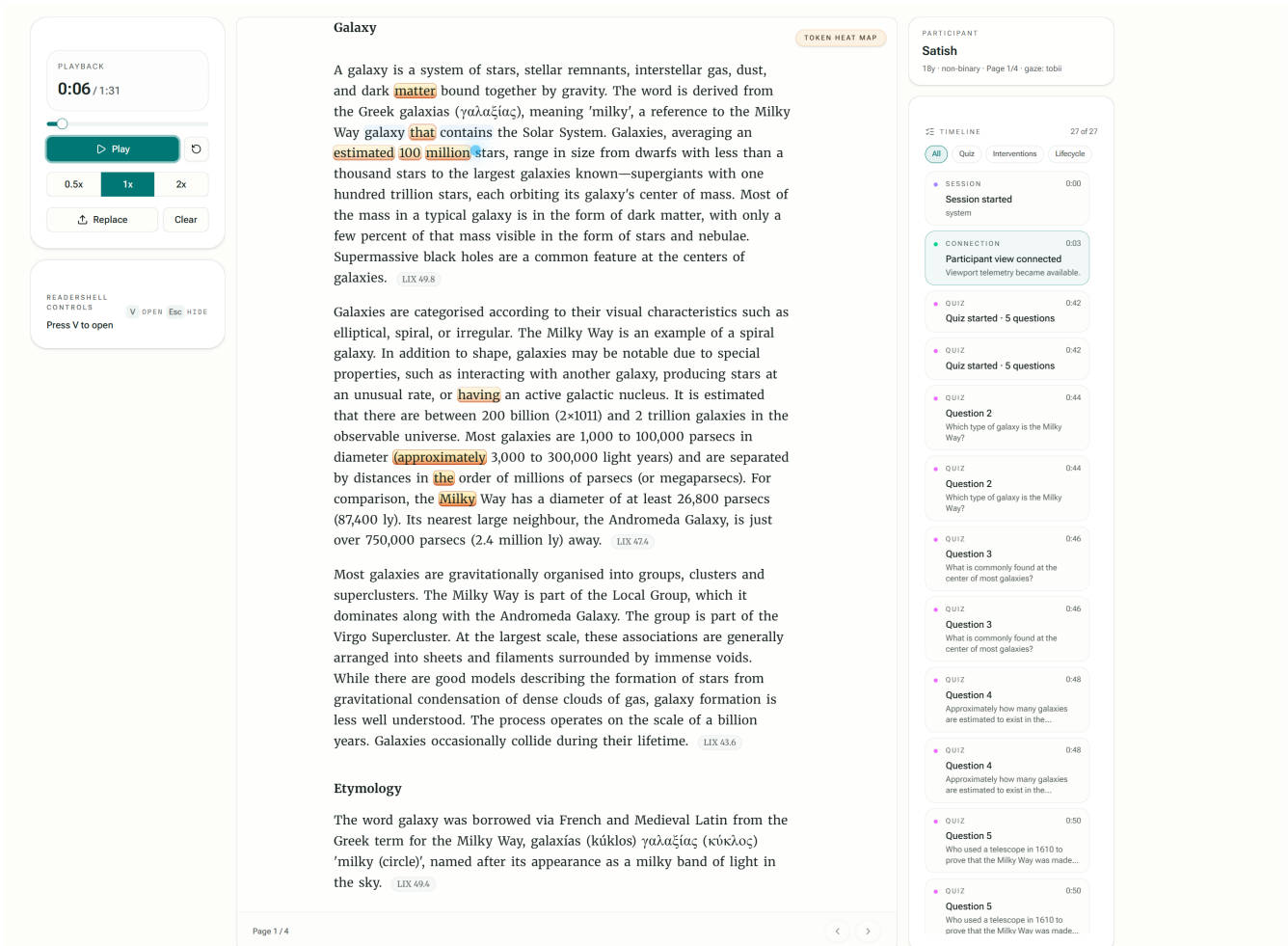


Figure 28: The replay player: playback controls on the left, the reconstructed reading view in the center, and the key-events timeline on the right.

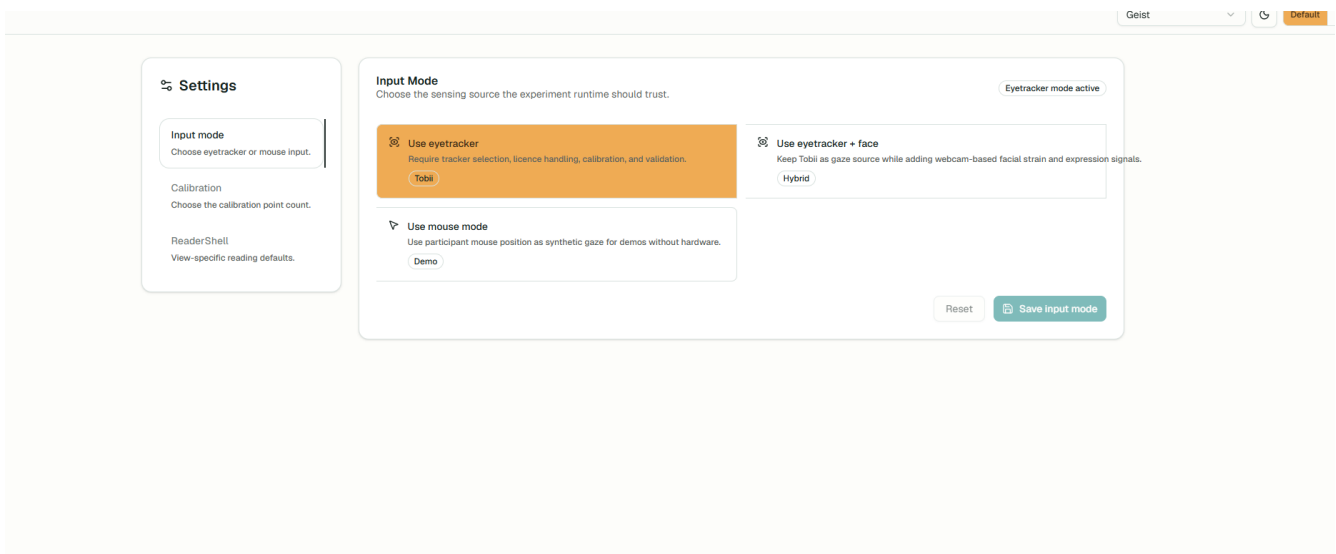


Figure 29: The Settings page with its section tabs (Input mode, Calibration, ReaderShell).

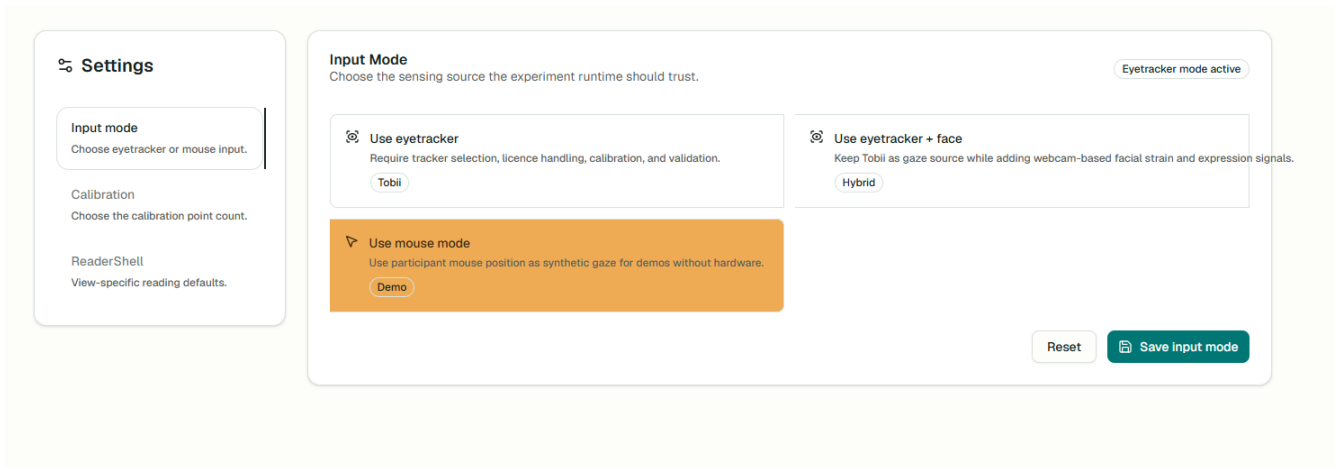


Figure 30: Input mode selection with Use mouse mode chosen.

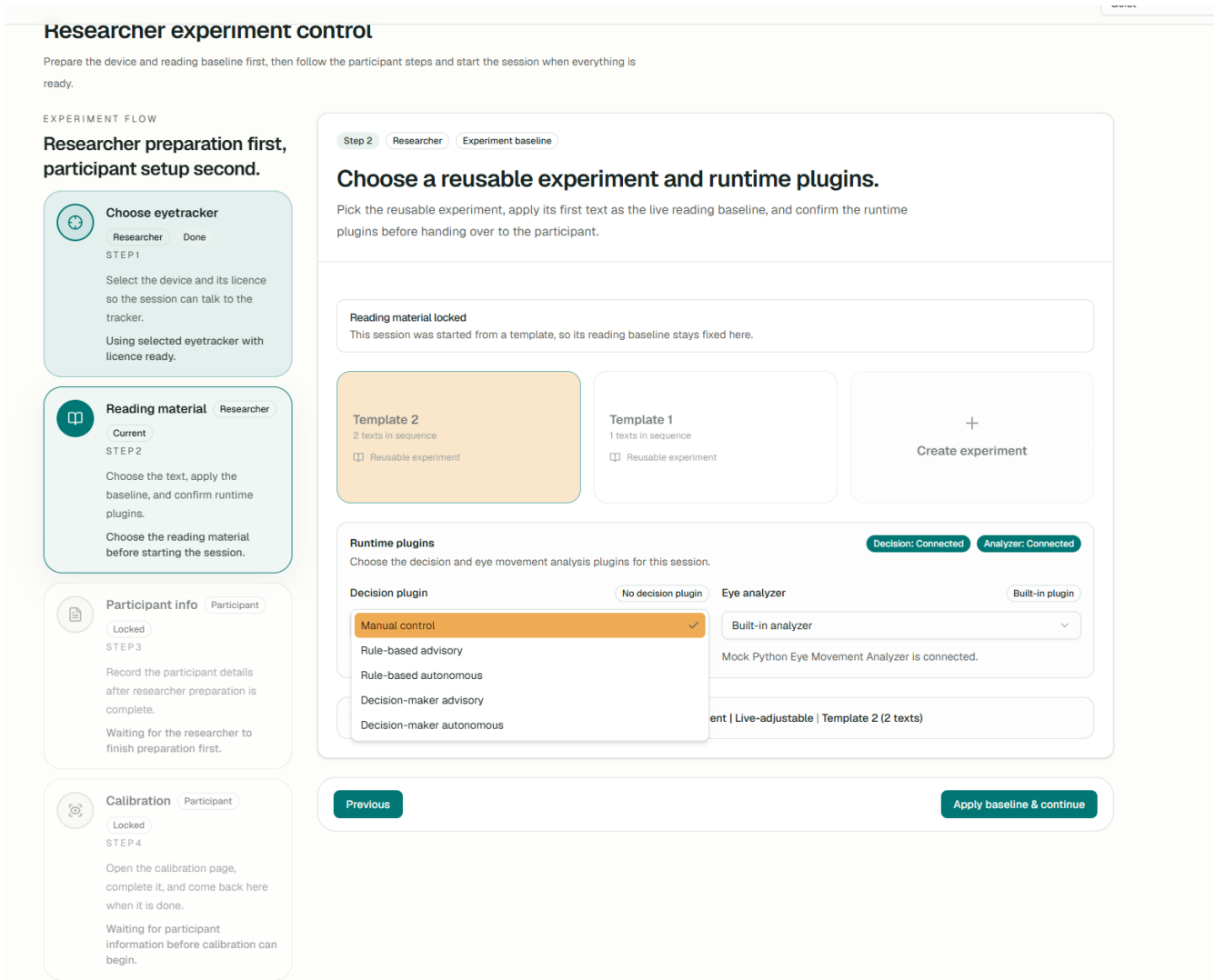


Figure 31: The runtime plugin selectors (decision plugin and eye analyzer) with their status badges.

9. Troubleshooting & FAQ

The participant page is stuck on “waiting”. The researcher hasn’t finished Steps 1–2 yet, or the session hasn’t started. Complete researcher preparation; the participant steps unlock automatically.

Calibration keeps getting interrupted. Calibration must stay in **full screen** and remain the visible tab for the whole run. If the browser blocks full screen, start again and allow it. Don’t switch tabs or windows during the run.

An external decision/analyzer option says “unavailable”. The external service isn’t connected (or doesn’t support that mode). Reconnect the service, or pick a built-in option.

The live mirror shows an “approximate” banner. The researcher view isn’t in full screen, or participant viewport data is missing. Click **Enter full screen** to restore an exact mirror.

I can’t start a template from Home. Only templates with status **Ready** appear under “Ready to run”. Open the template, set its status to **Ready** (it must contain at least one text), and save.

Can I run without hardware? Yes — switch **Settings** → **Input mode** to **Use mouse mode**.

What content formats are supported? Markdown only. PDF is intentionally not supported.

Where do my exports go? Downloads save to your browser’s download folder. **Saved** exports live inside the app and appear in **Replay**.

10. Glossary

Term	Definition
Reading material	One Markdown text + comprehension quiz + presentation defaults, saved for reuse.
Experiment template	A reusable, ordered sequence of texts with defaults, order mode, and a runtime strategy.
Baseline	The text + presentation condition applied at the start of the live reading session.
Live-adjustable vs Locked	Whether the researcher may change typography during the live session.
Sensing / input mode	The gaze source: eyetracker, eyetracker + face, or mouse.
Decision plugin	The component that decides whether/when to apply an intervention.
Eye analyzer	The component that derives fixation, saccade, and struggle state from gaze.
Intervention	A context-aware micro-change to the reading view (e.g. typography/appearance).
Validation	The post-calibration quality check measuring accuracy and precision.
Replay	Frame-by-frame playback of a recorded session.